Motor Safety Interlock with Multi-Sensor Check Using RTOS on LPC1768

Course: Operating Systems and Embedded Systems Design

Course Code: 25EECC304

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1 Problem Statement

Modern embedded and industrial systems require fail-safe motor operation to ensure safety under varying environmental conditions. In this project, a motor safety interlock system is designed such that a DC or stepper motor operates only when temperature, gas concentration, and obstacle distance are within predefined safe limits. If any one of these parameters becomes unsafe, the motor must stop immediately and the event must be logged. The system is implemented using a real-time operating system (RTOS) to ensure deterministic and reliable response.

2 Introduction

This project presents the design and implementation of a real-time motor safety interlock system using the LPC1768 ARM Cortex-M3 microcontroller and the RTX real-time operating system. Multiple sensors—LM35 temperature sensor, MQ-5 gas sensor, and HC-SR04 ultrasonic sensor—are interfaced with the microcontroller to continuously monitor environmental conditions. The motor is enabled only when all sensor readings are within safe thresholds.

The use of an RTOS enables task separation, immediate reaction to unsafe conditions, and structured inter-task communication using kernel objects such as semaphores, events, and message queues. Such systems are widely used in industrial automation, robotics, and safety-critical embedded applications.

3 Problem Analysis

3.1 Features Selected

- Multi-sensor safety monitoring
- Immediate motor shutdown on fault
- Event-based fault logging
- Deterministic task execution using RTOS
- Modular and scalable architecture

3.2 Assumptions Made

- Single motor control
- Stable sensor outputs
- External motor driver with separate power supply
- Pre-calibrated safety thresholds
- LPC1768 GPIO and ADC operate at 3.3 V logic

3.3 Peripherals Chosen

- HC-SR04 Ultrasonic Sensor for obstacle detection
- MQ-5 Gas Sensor for gas leakage detection
- LM35 Temperature Sensor for temperature monitoring
- DC Motor with two channel relay

- LPC1768 ADC for analog sensor acquisition
- GPIO for motor control and sensor triggering

3.4 Justification of Design Choices

The HC-SR04 sensor provides reliable distance measurement at low cost. The MQ5 sensor is commonly used for detecting combustible gases. The LM35 offers linear temperature output, simplifying ADC interpretation. RTX RTOS enables immediate response and structured synchronization, making it suitable for safety-critical systems.

4 System Architecture and Task Interaction

4.1 Block Diagram

The system consists of sensor acquisition tasks, a motor control task, and a logger task. Sensor data is stored in a shared structure protected by a semaphore. Events notify the motor task to evaluate safety conditions. Fault events are logged using a message queue.

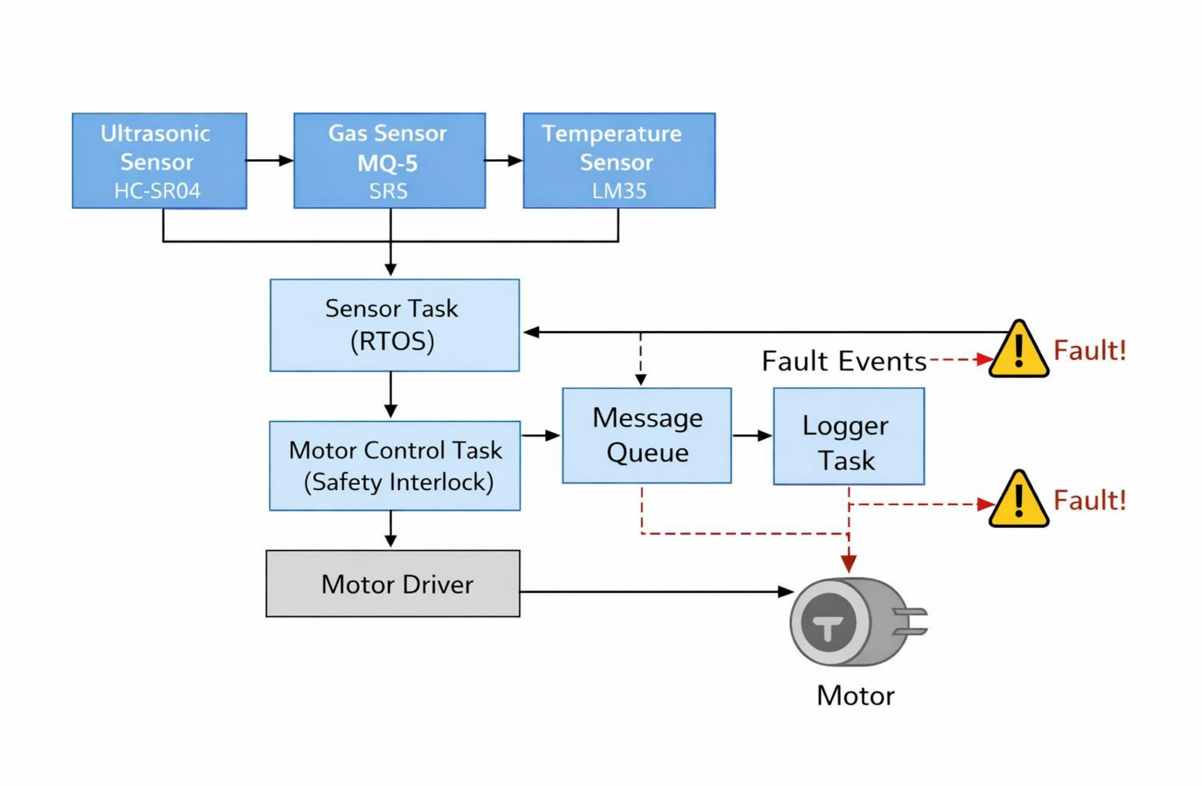

Figure 1: Block diagram of the motor safety interlock system using RTOS on LPC1768

4.2 RTOS Task Mapping

Task	Function	Priority

Sensor Task	Reads all sensors	Low
Motor Task	Safety evaluation and control	High
Logger Task	Event logging	Lowest

4.3 Sensor-Actuator Flow

1. Sensors acquire data periodically
2. Data stored in shared memory (semaphore protected)
3. Event signals motor control task
4. Motor task evaluates safety
5. Motor enabled or disabled
6. Fault event logged via message queue

5 Specifications and Calculations

5.1 HC-SR04 Ultrasonic Sensor

- Range: 2–400 cm
- Trigger pulse: 10 μs
- Distance calculation:

$$\text{Distance (cm)} = \frac{\text{Echo pulse width } (\mu s)}{58}$$

Safe distance threshold: 20 cm.

5.2 LM35 Temperature Sensor

- Output: 10 mV/$^\circ$C
- Operating range: 0–100 $^\circ$C • ADC resolution:

$$\text{ADC step} = \frac{3.3}{4096} \approx 0.805 \text{ mV}$$

$$T(^\circ C) \approx ADC \times 0.0805$$

Safe temperature threshold: approximately 38–40 $^{\circ}$C.

5.3 MQ-5 Gas Sensor

- Detects LPG, natural gas, hydrogen
- Analog output proportional to gas concentration
- Requires warm-up and calibration
- Threshold determined experimentally

5.4 Motor Specifications

- Type: DC motor
- Control: Enable pin via GPIO
- Power: External supply
- Safety: Disabled on any fault

5.5 Timing and Sampling

- Sensor sampling interval: 20 ms
- Motor response time: < 5 ms
- Event-driven scheduling (no polling delays)

6 Sensor Interfacing

6.1 LPC1768 Pin Connections

Signal	LPC1768 Pin
Ultrasonic TRIG	P0.26
Ultrasonic ECHO	P0.25
LM35 OUT	P0.23 (AD0.0)
MQ-5 AO	P0.24 (AD0.1)
DC Motor	P2.0 - P2.1
Ground	Common GND

6.2 Power Considerations

The LPC1768 Cortex-M3 board is powered through an external 5 V adapter. All the sensors are powered at 5 V from the board while the DC motor uses an external 12 V adapter. All grounds are common.

7 Results

7.1 Software Results

The RTOS successfully scheduled tasks with deterministic timing. The motor stopped immediately upon detection of any unsafe condition, and fault events were logged using the message queue.

7.2 Hardware Results

The system was tested on an LPC1768 development board. The motor operated only under safe conditions and stopped reliably for obstacle detection, gas leakage, or high temperature.

7.3 Implementation

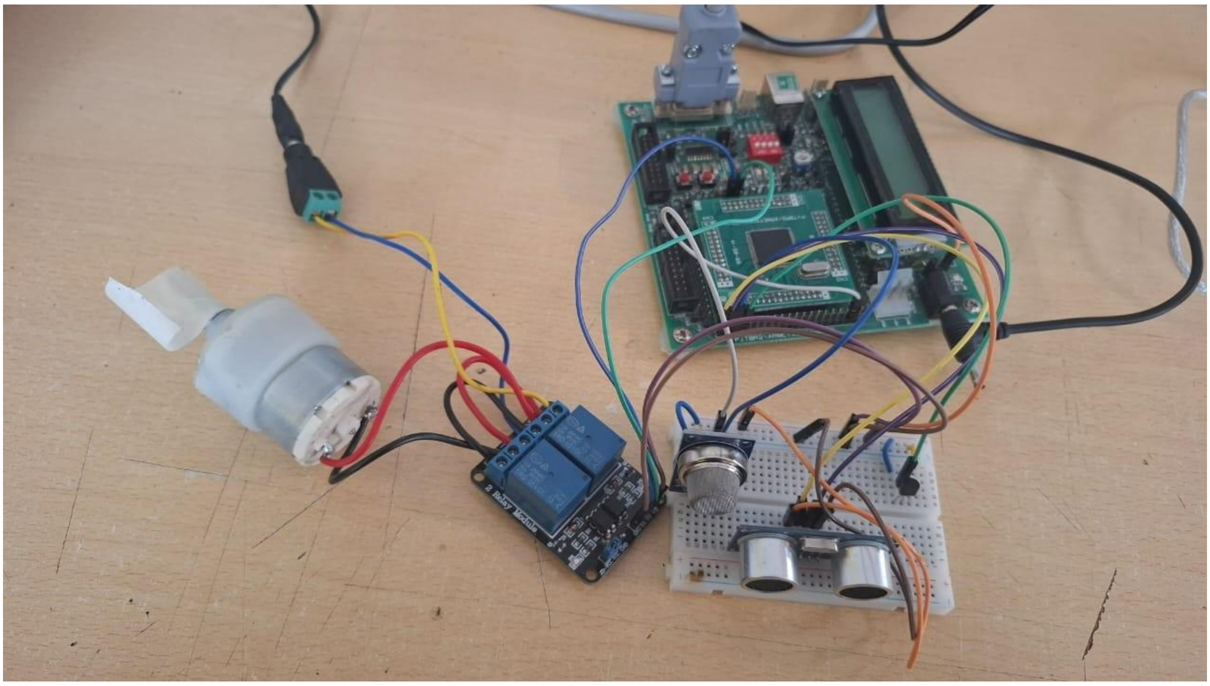

Figure 2: Block diagram of the motor safety interlock system using RTOS on LPC1768

Challenges Faced

- Calibration of MQ-5 gas sensor

- Noise handling in ADC readings
- RTOS configuration on LPC1768
- Ensuring immediate response without polling

8 Conclusion

The project successfully demonstrates a real-time motor safety interlock system using multiple sensors and RTOS principles. The motor operates only when all environmental parameters are safe and stops immediately upon detecting any unsafe condition. The use of RTX kernel objects such as events, semaphores, and message queues ensures reliable, deterministic, and scalable operation.

PPT Drive Link

https://docs.google.com/presentation/d/18Y9Cn0LBVDm16lc7tqJkduJ_9gQ0tUQr/edit?usp=sharing&ouid=113029078913291814704&rtpof=true&sd=true

Poster Drive link: https://drive.google.com/file/d/1nmA1C7CiTGGS-Gy9Ky8ayiNroBAMaJam/view?usp=sharing